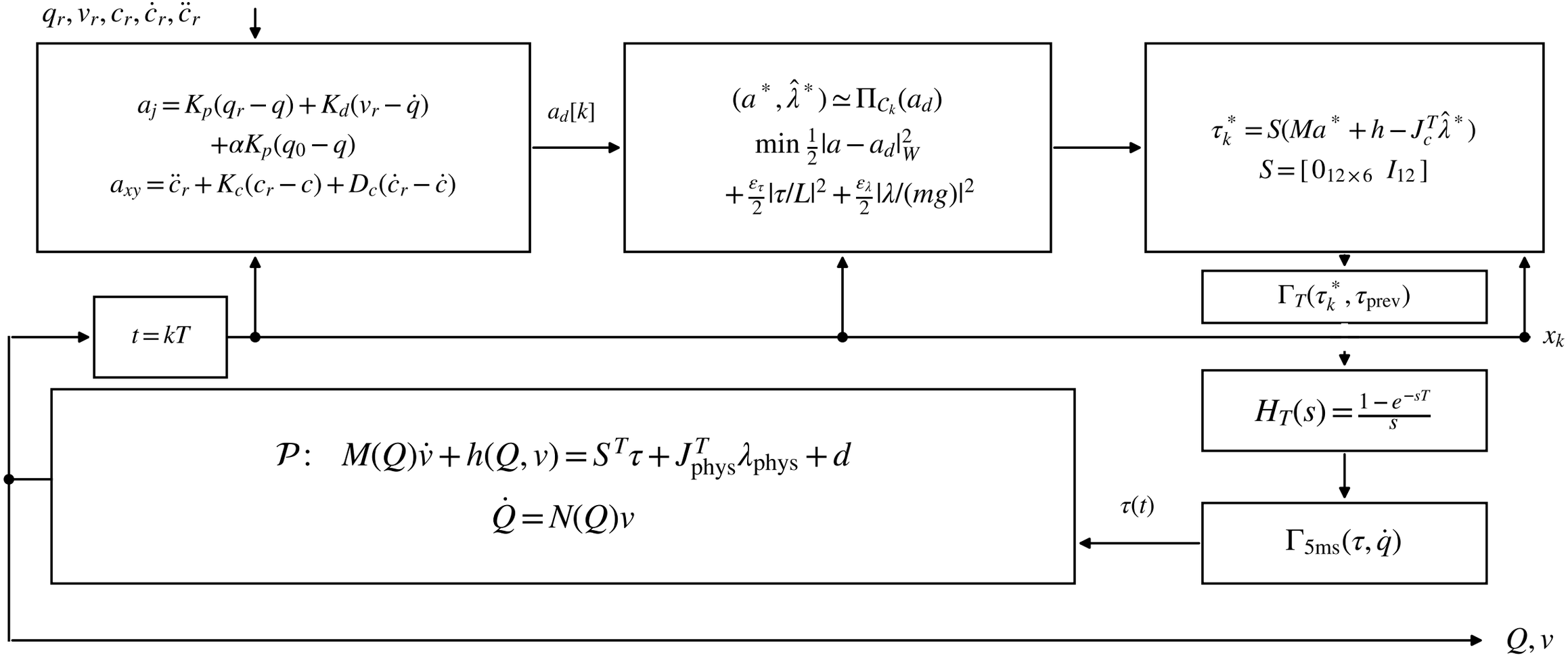
V16 | Actual nonlinear sampled control law

Mathematical blocks for the trained whole-body QP controller; Q has 19 coordinates, v has 18 components, tau has 12.



$$C_k:\ M_b a + h_b = J_{c,b}^T\lambda, \quad J_c a = a_c, \quad \tau_{\min} \le \tau \le \tau_{\max}$$

$$0 \le \lambda_z \le 500, \quad |\lambda_x|, |\lambda_y| \le \frac{0.6}{\sqrt{2}}\lambda_z, \quad \lambda_{\text{inactive}} = 0$$

IK/reference feedback and acceleration clipping are inside the sampled control law. QP uses 320 finite ADMM iterations.
Gamma_T: 20 ms slew clipping before hold. Gamma_5ms: speed/effort envelope after hold. Model and physical contact forces are distinct.